UTAC v2.0 Main Paper – Cover Sheet
Empirical Results, Separation from Theoretical Extensions, Reviewer Notes
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Note: This cover sheet supplements the unchanged UTAC v2.0 Main Paper. It summarizes the empirical findings and clearly delineates them from theoretical extensions, highlighting key sections for reviewer attention.

Empirical Core (Summary)

- **78 systems, clear β-clustering:**
  - Informational/SOC: $\beta \approx 4.5$
  - Biological: $\beta \approx 7.4$
  - Climate: $\beta \approx 11.0$
  - Neurodegenerative: $\beta \approx 13.0$
- **Statistical validation:** ANOVA and post-hoc tests demonstrate highly significant domain effects:
  - ANOVA: $F(4, 73) = 185.3$, $p < 10^{-20}$, $\eta^2 = 0.91$
  - Two-sample t-test (Info vs Others): $t(76) = 14.2$, $p < 10^{-20}$, Cohen's $d = 2.1$
- **Microfoundation:** β emerges from coupling-to-noise ratio J/T; mean-field-like, high-dimensional regimes occupy the lower β-band.

Separation from Theory Papers

- This **Main Paper** contains **empirical results only**.
- Companion papers (*Klimakaskade*, *Type-6*) are **discussion papers** and are clearly marked as such.
- **Empirical ≠ Theoretical:**
  - **Validated:** Domain-specific β-clustering (78 systems)
  - **Hypothetical:** Climate-seismic cascades, cubic-root jump dynamics

Reviewer Notes

We invite reviewers to pay particular attention to:

1. Robustness of Fits

- **Logistic vs. Power-Law:** Sigmoid (UTAC) fits vs. alternative models (power-law with cutoffs, MLE)
- **Bootstrap confidence intervals:** 1000 iterations, 95% CIs reported
- **Covariates:** Potential confounders (temporal resolution, measurement heterogeneity)

2. Sensitivity Near $R \approx \Theta$

- **Theoretical claim:** $\beta_{\text{dynamic}} \propto (R - \Theta)^{-1/3}$ (discussed in theory papers)
- **Empirical question:** Does this form improve fits for systems near threshold?
- **Validation approach:** Requires separate datasets with high temporal resolution near Θ

3. Domain Classification

- **Criteria:** Physical substrate, timescale, dimensionality, coupling type
- **Ambiguous cases:** Some systems (e.g., epidemics) could classify as Informational or Biological
- **Sensitivity analysis:** Re-classification does not affect main conclusions (tested)

4. Golden Ratio Scaling

- **Empirical observation:** $\beta_n \approx \Phi^{n/3}$ predicts domain means with ¿8% error
- **Mechanistic question:** Why Golden Ratio? (Geometric interpretation provided; empirical validation ongoing)
- **Falsification:** If β-values systematically deviate ¿20% from $\Phi^{n/3}$ in future datasets

Key Strengths

1. **Large sample size:** 78 systems across 5 domains (vs. 36 in v1.0)
2. **Statistical rigor:** Effect sizes reported ($\eta^2 = 0.91$, Cohen's $d = 2.1$)
3. **Clear separation:** Empirical claims (this paper) vs. theoretical extensions (companion papers)
4. **Reproducibility:** Full dataset, code, and documentation on Zenodo

Known Limitations

1. **Publication bias:** Threshold systems more likely to be published

2. **Measurement heterogeneity:** Different domains use different techniques
3. **Sample size:** 78 systems is robust but could expand to 100-200
4. **Cross-domain coupling:** Not yet empirically validated (theoretical extension only)

Main Paper: Domain-Specific Universality in Threshold Activation Criticality
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